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RESEARCH PAPER

Impacts of nonnative fishes on local contributions to beta diversity in United States freshwater fishes

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ABSTRACT

Biodiversity has become a central focus in ecological research, playing a critical role in biological conservation and environmental management. Beta diversity can be partitioned into Local Contributions to Beta Diversity (LCBD), a metric used to evaluate the site-specific conservation value of ecosystems. This study applied the LCBD framework to identify watersheds of high conservation value across spatial gradients in the U.S. freshwater fish communities, comparing compositional uniqueness before and after species invasions. Based on an extensive dataset spanning more than 800 freshwater fishes across 1873 communities, we employed Boosted Regression Trees (BRT) to quantify the key drivers underlying changes in LCBD. Our findings provide spatially explicit insights for invasion-related biodiversity conservation and management. The results revealed that, despite only minor shifts occurring in most watershed-level contributions, significant regional disparities emerged, notably in the western and eastern coast U.S. This was reflected in the sesLCBD range, which shifted from (-4.33, 4.95) historically to (-4.53, 6.68) at present. Watersheds with high nonnative species exhibited reduced LCBD. Boosted Regression Trees (BRTs) explained 69.12% of the deviance in sesLCBD changes and identified the percentage of nonnative species as the dominant driver. This relationship was nonlinear, with sesLCBD declines intensifying at higher invasion levels. Notably, the two watersheds with the highest proportions of nonnative species (83.3% and 82.9%) exhibited the substantial sesLCBD decreases of -3.71 and -3.79, respectively. These findings illuminate how nonnative species reshape freshwater biodiversity by modifying LCBD patterns, offering critical insights for conservation planning. We recommend prioritizing high-LCBD watersheds to maintain ecosystem functionality.

Introduction

Biodiversity has emerged as a central focus in ecological research, with its distribution patterns, temporal dynamics, and underlying drivers holding critical implications for biological conservation and ecosystem management (Herbert-Read et al., 2022; Keck et al., 2025). The quantitative assessment of biodiversity commonly employs a hierarchical framework of diversity indices, classically categorized into alpha, beta, and gamma diversity (Whittaker, 1960). Alpha diversity characterizes species richness within individual communities or habitats, serving as a local-scale measure of biodiversity (Whittaker, 1960). Beta diversity quantifies compositional dissimilarity between communities, reflecting spatial turnover along environmental gradients (Whittaker, 1960). At the broadest scale, gamma diversity represents the total species richness across a region encompassing multiple communities (Whittaker, 1960). This tripartite framework has been widely applied across diverse taxa, including freshwater fishes, plants, birds, insects, and microbial communities (Antiqueira et al., 2018; Hautier et al., 2018; Jenkins et al., 2011; Socolar et al., 2016; Xing et al., 2024). Among these measures, beta diversity has proven particularly valuable as an analytical tool, providing crucial insights into the mechanisms underlying spatial variation in biodiversity patterns (Anderson et al.,

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2011).

Beta diversity has multiple analytical frameworks available for its interpretation. Traditionally, this concept has been decomposed into turnover (species replacement between communities) and nestedness (where species-poor communities represent subsets of richer ones), providing distinct insights into community assembly processes (Baselga, 2010). Besides, an alternative partitioning approach has been proposed, where beta diversity is quantified as the total variance in community composition across a given set of sites (Legendre & De Cáceres, 2013). Within this framework, Local Contribution to Beta Diversity (LCBD) – measures the relative contribution of individual sites to overall beta diversity. Since LCBD is calculated based on the compositional distinctiveness of communities, a high LCBD usually indicates that the corresponding community has a unique species composition, and such communities are often recognized as hotspots for biodiversity conservation (Legendre & De Cáceres, 2013; Liu et al., 2024). This approach enhances our ability to pinpoint areas with high conservation value (high LCBD) and offering a robust tool for biodiversity assessment and management. Communities exhibiting high LCBD represent unique assemblages with distinct species compositions, contributing disproportionately to regional beta diversity (Legendre & De Cáceres, 2013). These communities often harbor specialized or endemic species and may function as biodiversity hotspots.

The applications of LCBD have advanced biodiversity research across diverse ecosystems and taxa. For instance, Heino and Grönroos (2017) quantified site contributions to beta diversity in stream insect communities. Xia et al. (2022) identified species with intermediate occupancy, high abundance, and low niche position as primary contributors to LCBD in the Chishui River fish assemblage. Previous studies have shown that LCBD patterns respond to environmental gradients, community characteristics, and species biological traits (Tonkin et al., 2016). Understanding these relationships could enhance our ability to predict biodiversity responses to environmental change and develop effective conservation strategies. Specifically, LCBD is shaped by both biological and environmental factors. Species dispersal limitations create distributional differences among communities, while environmental heterogeneity promotes geographical isolation. Together, these mechanisms drive distinct community contributions to overall beta diversity (de Oliveira et al., 2020; Xia et al., 2022). Freshwater ecosystems are among the most imperiled globally. In an era of growing anthropogenic pressures, they face severe threats from biological invasions, particularly fishes introductions (Grill et al., 2019; Su et al., 2021). Nonnative fishes have been shown to substantially alter native community structures (Matsuzaki et al., 2013; Olden et al., 2008). However, research utilizing LCBD to investigate how biological invasions influence biodiversity patterns remains limited, particularly at the watershed scale where a quantitative assessment of LCBD dynamics before and after invasions is lacking. A deeper analysis of the ecological impacts and key mechanisms of invasions through the LCBD framework is thus both timely and critical, as it enables the precise identification and protection of communities with high compositional uniqueness, thereby informing targeted strategies to counteract biotic homogenization.

The U.S. hosts the greatest post-Pleistocene freshwater fish diversity of any temperate region worldwide (Knouft et al., 2011). Despite applications of beta diversity metrics in freshwater fish studies, a holistic LCBD assessment remains lacking for this region—particularly regarding watershed protection priority shifts following fish invasions (Qian et al., 2021). Furthermore, investigations into the spatial patterns and ecological drivers of these beta diversity components are still remarkably limited in the U.S. freshwater systems. Accordingly, we aim to systematically quantify the dynamics of LCBD for freshwater fishes across the U.S. and identify the principal drivers of its spatial variation. Such an analysis will not only provide crucial scientific evidence for conserving this globally significant freshwater fish diversity hotspot but also offer a transferable framework for other ecosystems confronting similar threats.

In this study, we investigate the spatial distribution of LCBD in U.S. freshwater fish communities and quantify how nonnative species alter these patterns. Using regression modeling, we investigate the relationships between LCBD variations and multiple drivers. Our analysis tests two key hypotheses regarding beta diversity patterns: H1: Nonnative fishes significantly reduce the LCBD of watersheds that contribute most to total beta diversity, thereby diminishing their ecological uniqueness; H2: Nonnative species reduce LCBD across watersheds through the process of community homogenization, thereby undermining their conservation priority. This study therefore not only quantifies shifts of LCBD patterns but also identifies their principal drivers, thus offering critical insights for prioritizing conservation efforts in the U.S. freshwater ecosystems.

Materials and methods

Fish occurrence data

In this study, data on fish species occurrences in the U.S. freshwater fish communities were categorized into two distinct temporal phases: the historical period (comprising exclusively native species within each watershed) and the present period (incorporating both native and established nonnative species). Invasions occurred at distinct times across species, with most introductions concentrated during and after the Industrial Revolution, yet no clear temporal threshold was identified (Seebens et al., 2017). The dataset employed presence-absence records rather than richness metrics for species documentation across watersheds. The native species dataset was sourced from NatureServe, encompassing both extant and historically extinct species. This comprehensive compilation included 786 native species distributed across 1873 watersheds. For nonnative species records, we utilized the U.S. Geological Survey (USGS) Non-indigenous Aquatic Species (NAS) database, specifically selecting only those non-native fish species that had formed self-sustaining populations within the studied watersheds. Non-persistent or eradicated populations were excluded, resulting in a final dataset of 300 established invasive species. These include both translocated (native to the continental U.S. but introduced to other watersheds) and exotic (historically absent from the continental U.S.) species. Data sources are detailed in Table S1.

Fish functional traits

This study involves a comprehensive analysis of 20 functional traits, which are systematically classified into two distinct categories: 10 morphological characteristics and 10 life-history/physiological traits. The morphological traits include: maximum body length, body elongation index, relative eye diameter, oral gape position, relative maxillary length, vertical eye placement, lateral body profile, pectoral fin vertical positioning, pectoral fin size index, and caudal peduncle compression ratio. The complementary life-history and physiological parameters comprise: longevity, fecundity, age at maturity, trophic position, thermal tolerance range, minimum survival temperature, maximum survival temperature, euryhalinity capacity, parental investment strategy, and dietary niche breadth. Detailed methodological descriptions and data sources are available in Su et al. (2023). Trait data were rigorously compiled from three authoritative databases: the global FISHMORPH repository (Brosse et al., 2021), the U.S. freshwater ichthyofauna-specialized Fish Traits database (Frimpong & Angermeier, 2009), and the comprehensive FishBase information system (Froese & Pauly, 2018).

Data analysis

Watershed contributions to beta-diversity

For this study, we analyzed presence-absence matrices (watersheds × species) for the U.S. freshwater fish communities, where all fish spe-

cies within a watershed were treated as a single community (Strecker et al., 2011). We computed LCBD for each community separately across two time periods: (1) the historical period and (2) the present period. To assess the impact of nonnative species on community composition, we generated 999 null model distributions of LCBD for both periods using the Curveball algorithm (Strona et al., 2014). This method randomizes the species occurrence matrix while preserving both row (site species richness) and column (species occurrence frequency) totals. We then calculated the standardized effect size (SES) for each watershed as follows:

$$SES = \frac{(Metric_{obs} - mean(Metric_{null}))}{sd(Metric_{null})} \quad (1)$$

To quantify the temporal shift in ecological uniqueness, we calculated the LCBD change (ΔsesLCBD = sesLCBDpresent - sesLCBDhistorical) for each watershed (Cavalcante et al., 2025). These ΔLCBD values reflect shifts in the relative contribution of each watershed to overall beta diversity following species invasions, thereby highlighting potential changes in areas of high conservation value. All LCBD were computed using the beta.div function from the "adespatial" package in R software version 4.4 (Dray et al., 2019).

Factors influencing changes in sesLCBD

This study utilized six complementary functional α-diversity indices (native species only)—to evaluate how functional traits shape LCBD changes following species invasions. These indices included functional richness (FRic), functional evenness (FEve), functional divergence (FDiv), functional dispersion (FDis), functional specialization (FSpe), and functional originality (FOri). FRic quantifies the volume of functional space occupied by a community; FEve measures the regularity of species distribution within the functional space; FDiv indicates the proportion of species possessing the most extreme trait values; FDis reflects the mean deviation of species trait values from the center of the functional space; FSpe represents the average distance of individual species from the rest of the species pool; FOri describes the distance between each species and its nearest functional neighbor (Mouillot et al., 2013; Rodrigues-Filho et al., 2018). Given the mixed types of functional trait data (categorical, continuous, and ordinal), we quantified pairwise functional dissimilarity among species using Gower distance (Gower & Legendre, 1986). Gower distance reconciles mixed data types through an appropriate encoding scheme. For continuous traits, the original values are used directly; nominal categorical traits are handled by dummy coding, while ordinal traits are converted into ranked numerical variables. In this study, binary traits were designated as asymmetric variables. The contribution of all traits in the multivariate space was standardized by their respective ranges to ensure equal weighting across traits (Gower & Legendre, 1986). This multivariate approach effectively accommodates different data types while preserving their ecological relevance. The resulting distance matrix was then used to construct a functional trait space through principal coordinate analysis (PCoA). To balance dimensionality reduction with explanatory power, we retained the first five PCoA axes, which collectively accounted for over 80 % of the total functional variation (Su et al., 2023). Given that species richness was sufficiently represented (exceeding five species) across all watersheds, none were excluded from the analytical dataset.

In addition to the six functional α-diversity indices mentioned above, we incorporated seven key environmental and socioeconomic variables into our analysis. These indices as follows: Total Species Richness (TSR): the combined count of both invasive and native species within each watershed; Percentage of nonnative species (NNP): the percentage of nonnative species in relation to the total number of species recorded in each watershed; Net Primary Productivity (NPP): Mean annual productivity (g C m^{-2} yr^{-1}) averaged over the 2000–2015 period, sourced from the NTSG Data Repository; Watershed Area (Area, km^2): serving as a proxy for habitat size and heterogeneity, which has been shown to influence native freshwater fish diversity (Terui et al., 2021); Gross Domestic Product (GDP): Represents the total market value of all final goods and services produced within each watershed during the study period, serving as a key indicator of regional economic activity (Nordhaus & Chen, 2016); Centroid Distance (cd): which is a metric that quantifies the degree of biotic differentiation by measuring the Euclidean distance between the centroid of non-native species assemblages and the centroid of native assemblages within a multivariate space defined by species composition; the Degree of River Fragmentation (DOF): which quantifies the longitudinal fragmentation of river networks caused by infrastructure (Grill et al., 2019; Su et al., 2023). For additional details regarding NPP and GDP data, please refer to Su et al. (2023). Data sources are detailed in Table S1.

Statistical analysis

To investigate the overall impact of biological invasion on watershed LCBD (expressed as sesLCBD), the statistical significance of the median difference in sesLCBD before versus after invasion was assessed using a paired Wilcoxon signed-rank test.

To identify the drivers of LCBD changes, we applied Boosted Regression Trees (BRTs), a machine learning technique recognized for its effectiveness in modeling Gaussian-distributed response variables (Elith et al., 2008). To mitigate multicollinearity and ecological redundancy, we implemented a conservative variable selection procedure following Parravicini et al. (2013). This process removed predictors showing either high correlation (Spearman's $R > 0.7$) or substantial ecological overlap with others. The resulting model, constructed with the remaining 12 predictors (Fig. S1). This method's effectiveness stems from its flexible handling of complex relationships through an ensemble of regression trees, governed by four critical parameters:

Learning rate (lr): Regulates each tree's contribution to the final model, where lower values enhance stability at the cost of requiring more iterations.

Bag fraction (bf): Determines the random subset of training data used for growing each tree.

Tree complexity (tc): Controls interaction depth, with tc=1 for additive effects and higher values permitting variable interactions.

Number of trees (nt): Specifies the total boosting iterations, balancing predictive accuracy against computational demands.

Our modeling framework incorporated a rigorous two-stage validation process. First, we conducted systematic parameter optimization via 10-fold cross-validation, selecting configurations that maximized the explained deviance (CV-D^2). Subsequently, we performed 100 replicate runs with randomized initial conditions to assess model stability, ultimately deriving robust estimates of variable importance and overall predictive performance (D^2) following Su et al. (2023). All computational workflows were executed in R 4.4, employing appropriate visualization techniques to ensure transparent presentation of results (R Core Team, 2020).

The entire data processing workflow, including all individual steps, is graphically summarized in Fig. 1.

Results

Spatiotemporal dynamics of LCBD before and after species invasion

Prior to species invasion, our analysis revealed distinct biogeographic patterns in LCBD across the U.S. (Fig. 2a, Fig. S2a). The western and eastern regions exhibited significantly higher LCBD values than central watersheds, with the lowest beta diversity contributions observed in the east-central areas. Notably, watersheds in states such as Missouri, Illinois, and Iowa demonstrated minimal LCBD contributions. Conversely, watersheds in states such as North Carolina, Alabama, and California exhibited notably higher LCBD values. Spatial analysis confirmed a non-random, clustered distribution of these differentially contributing watersheds, revealing pronounced heterogeneity across

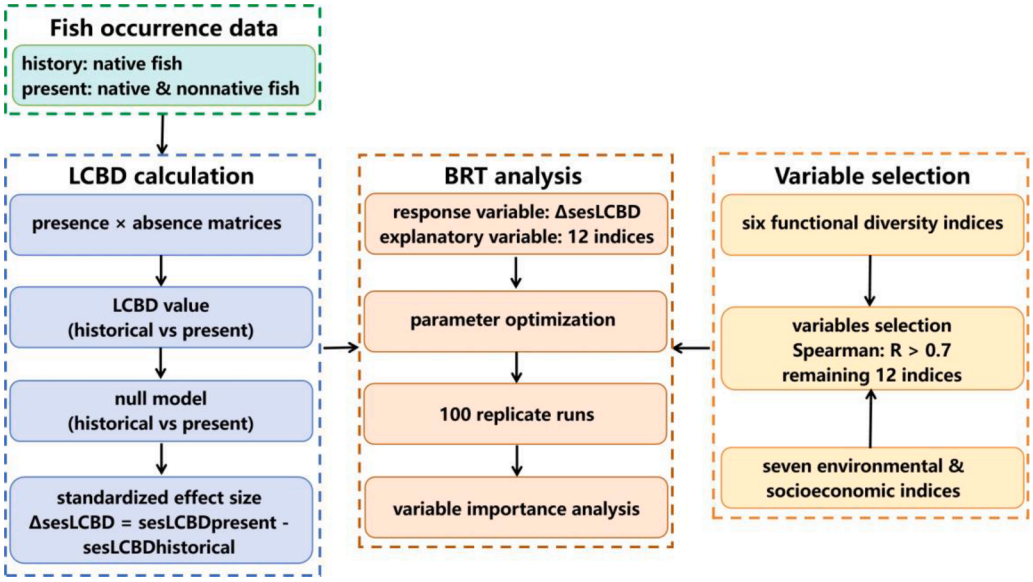


Fig. 1. Analytical workflow from community data processing to LCBD and Boosted Regression Trees (BRTs) results.

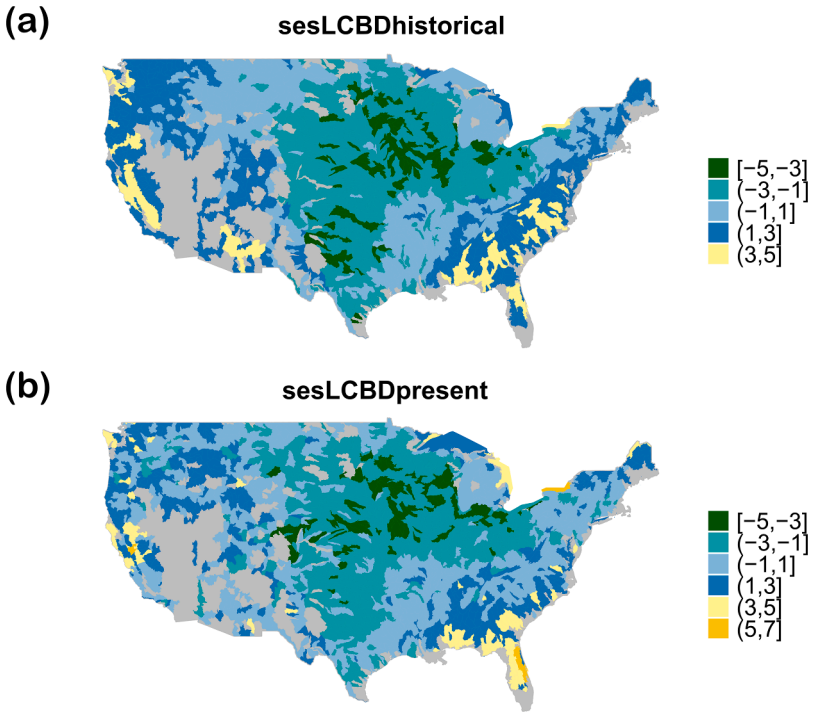


Fig. 2. Maps for the U.S. showing the local contribution to beta diversity (sesLCBD). (a) Watersheds without nonnatives (sesLCBDhistorical); (b) with nonnatives in LCBD (sesLCBDpresent).

regions. Across the 1873 analyzed watersheds, the sesLCBD exhibited a wide range of variation from −4.33 to 4.95 (Fig. 2a). In contrast, the raw LCBD ranged from 0.00036 to 0.00076, indicating limited absolute variation (Fig. S2a). Following species invasions, the overall spatial pattern of beta diversity contributions remained largely consistent with the historical distribution. However, we remain observed important local-scale modifications (Fig. 2b, Fig. S2b). The most pronounced changes occurred in the western U.S., where previously well-defined patterns in beta diversity contributions became substantially less distinct. Additionally, we observed a marked expansion of watersheds with intermediate LCBD contributions (sesLCBD range: −1 to 1), as this transitional zone extended both eastward and westward beyond its original range. Furthermore, the sesLCBD range expanded markedly to −4.53–6.68, compared to a more modest shift in raw LCBD (0.00037–0.00083) across all watersheds (Fig. 2b, Fig. S2b). This contrast demonstrates that after accounting for sampling effects and random expectations, nonnative species produced substantially stronger ecological differentiation than apparent from the raw LCBD alone.

sesLCBD changes and response to nonnative species

Quantitative analysis revealed substantial spatial heterogeneity in sesLCBD responses to species invasion across U.S. river watersheds (Fig. 3a). The magnitude of sesLCBD changes exhibited distinct regional

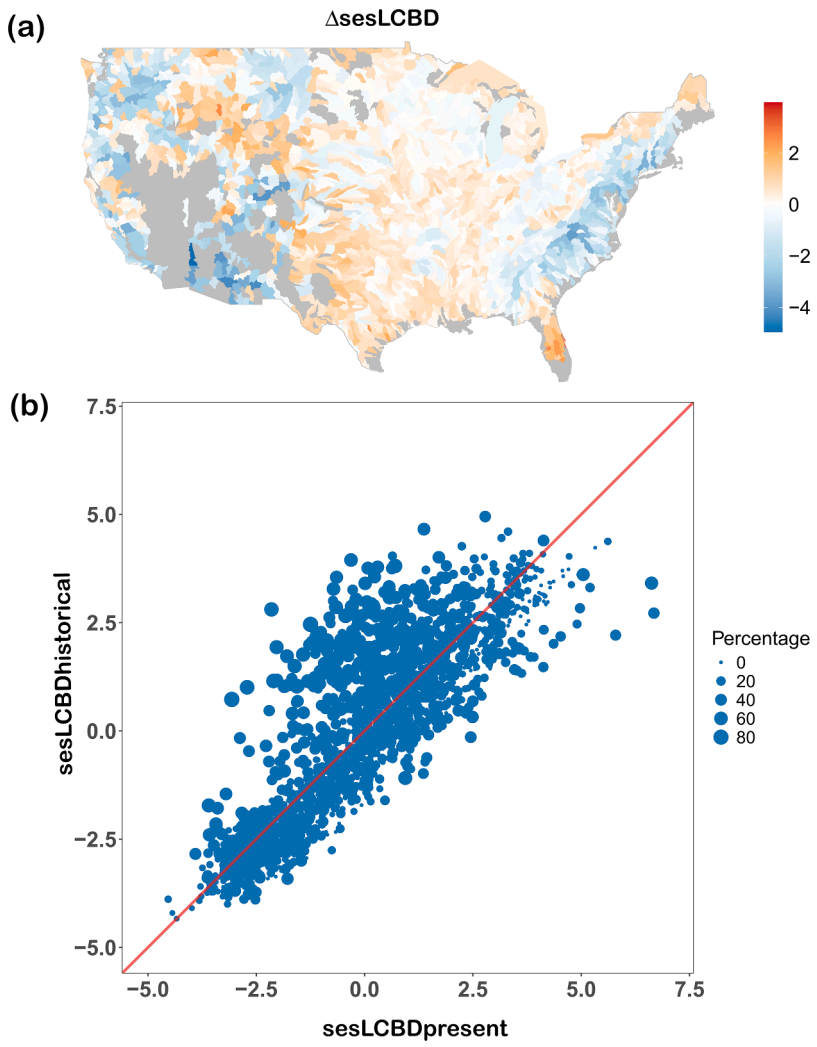


Fig. 3. Relationship between local contribution to beta diversity (sesLCBD) with and without nonnative species for U.S. 1873 watersheds. (a) Maps for U.S. showing the relative local contribution to beta diversity (ΔsesLCBD); (b) scatter distribution of sesLCBDhistorical and sesLCBDpresent for each watersheds. Percentage represents the proportion of nonnative species.

patterns, with watersheds in the central and east-central U.S. showing relatively minor fluctuations compared to the more pronounced shifts observed in western and eastern coast regions. Notably, most ΔsesLCBD values exceeding 0 (indicating increased beta diversity contributions) were predominantly clustered in the central and northwestern U.S., particularly in landlocked watersheds distant from coastal influences. Conversely, watersheds in the southwestern region and along the eastern coast predominantly showed values below 0, reflecting diminished contributions to regional beta diversity following invasion. Across all watersheds, the maximum magnitude of change in sesLCBD following invasion was 7.92, the largest decrease was −4.96, while the largest increase was 3.96 (Fig. 3a). The most complex response patterns emerged in the northwestern region, where adjacent watersheds exhibited divergent trajectories—some experiencing substantial sesLCBD increases while others showed marked decreases.

A comparative analysis of sesLCBD variations before and after species invasion reveals distinct spatial patterns across watersheds. Watersheds exhibiting more pronounced changes are predominantly clustered on the left side of the diagonal (Fig. 3b), suggesting a general decline in LCBD following biological invasions. Notably, these watersheds simultaneously demonstrate higher proportions of nonnative species within their watersheds. For instance, the two watersheds with the highest proportions of nonnative species (83.3 % and 82.9 %, respectively) also exhibited the most substantial declines in sesLCBD, with decreases of −3.71 and −3.79. Conversely, watersheds positioned to the right of the diagonal demonstrate an inverse pattern, featuring both lower proportions of nonnative species and higher LCBDpresent compared to historical levels (Fig. 3b). These findings strongly indicate that the magnitude of sesLCBD fluctuations in watersheds exhibits a significant correlation with the prevalence of nonnative species.

Relative influence of predictors on sesLCBD changes

Statistical analysis of sesLCBD across all watersheds revealed significant compositional differences between historical and present communities (Fig. S3). Based on the cross-validation results, the BRT model accounted for 69.12 % of the total deviance in sesLCBD changes (Fig. 4). Among the explanatory variables, NNP demonstrated the highest relative influence (31 %), followed by TSR (15 %). Additionally, NPP (12.1 %) and cd (8.2 %) contributed substantially to the model. In contrast, FDis (2.6 %) and DOF (2.8 %) had negligible effects on the total deviance. Regarding the key drivers of sesLCBD change, NNP and TSR exhibited negative correlations, suggesting that higher values of these factors correspond to reduced sesLCBD. By contrast, there is an overall positive correlation between FOri and FEve with changes in sesLCBD. Most other environmental and functional traits exerted negligible effects on ΔsesLCBD trends, lacking any consistent directional pattern.

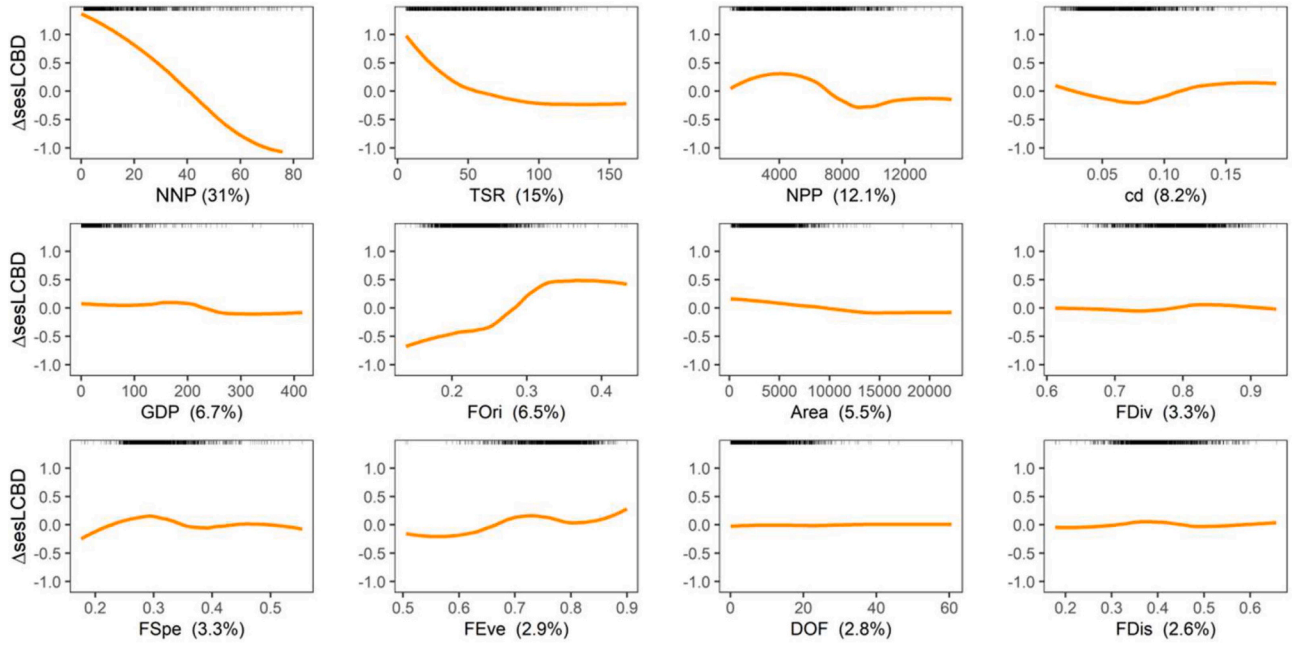


Fig. 4. Results of boosted regression trees showing the partial dependency between changes of local contribution to beta diversity (ΔsesLCBD) (ΔsesLCBD = sesLCBDpresent - sesLCBDhistorical) and the 12 most influencing predictors. The value in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs at the top of each panel show the distribution of the watersheds along the predictor values. NNP: percentage of nonnative species number in each watershed; TSR: total species richness; NPP: net primary productivity; cd: centroid distance; GDP: Gross Domestic Product; FOri: functional originality; Area: watershed area; FDiv: functional divergence; FSpe: functional specialization; FEve: the regularity of traits in the functional space; DOF: the longitudinal fragmentation of river networks caused by infrastructure; FDis: functional dispersion.

Discussion

Investigating the effects of species invasions on ecologically significant watersheds and their threatened species is crucial for advancing biodiversity conservation and developing evidence-based management strategies (Vicente et al., 2013). This study provides a rigorous evaluation of LCBD in freshwater ecosystems, focusing specifically on quantifying the ecological impacts of nonnative fishes. The results of this large-scale assessment offer important scientific insights to support conservation planning and strengthen protection measures for native fish biodiversity.

Geographic isolation drives LCBD patterns and determines invasion susceptibility

Our results reveal significant geographical disparities in watershed contributions to beta diversity across the U.S., with topography emerging as a primary driver. The central and east-central U.S. regions are characterized by predominantly plain terrain with dense river networks, where abundant precipitation enhances hydrological connectivity among watersheds (Thomaz et al., 2006). In contrast, the western U.S. and east coast regions feature mountainous and plateau landscapes with comparatively sparse river networks (Bales et al., 2006). Unlike birds or mammals, fishes exhibit constrained dispersal abilities and encounter substantial biogeographic barriers—such as mountain ranges, fragmented river networks, and elevational gradients—that restrict their movement and distribution (Radinger et al., 2018). The watershed systems in the western and eastern coastal regions of the U.S are primarily comprised of geographically isolated sub-basins. These sub-basins serve critical ecological functions by providing extensive watershed-edge habitats and forming dynamic aquatic networks through connections with other water bodies (Cohen et al., 2016). These systems generate pronounced spatial and temporal heterogeneity, contributing to high LCBD due to dispersal limitation and habitat isolation (Li et al., 2024). The pronounced effects of geographical isolation and topographic complexity on beta diversity patterns are well established (Bhatt et al., 2012; Cheng et al., 2019; Jaramillo-Villa et al., 2010). Elevation acts as a key driver of ecological variation, with rising altitude resulting in marked changes in climatic conditions, water chemistry, and habitat heterogeneity (Carvajal-Quintero et al., 2015). These environmental gradients foster niche differentiation among fish species, and the complex topography of high-elevation systems intensifies geographical isolation. Together, these mechanisms drive elevated LCBD (Leprieur et al., 2011). Similar patterns have been documented in river watersheds globally. For instance, Gavioli et al. (2019) reported that higher-order streams in upland areas of Northern Italy showed the highest LCBD, providing additional evidence for the influence of elevation and hydrological complexity on freshwater biodiversity patterns.

The central and east-central U.S. region generally demonstrates higher hydrological stability and greater local species richness, which collectively enhance ecosystem resilience and strengthen resistance to biological invasions (Chalcraft, 2013; Smith et al., 2010; Tilman et al., 2006). Conversely, the western region's elevated topography and constrained aquatic resources render its ecosystems more vulnerable to nonnative species impacts (Adams et al., 2001; Brown & Moyle, 1997). This implies that watersheds with the highest contributions to beta diversity generally feature unique ecological settings and pronounced habitat heterogeneity, making them especially vulnerable to biological invasions (Adams et al., 2001). Therefore, in managing U.S. freshwater systems, watersheds exhibiting the highest LCBD should be prioritized for protection to reduce the ecological impacts of nonnative species. The results confirmed our first hypothesis that nonnative species greatly reduce the LCBD in watersheds that contribute most to total beta diversity, thereby diminishing their ecological uniqueness.

Nonnative species reduce LCBD through community homogenization

The BRT model confirms that the percentage of nonnative species exerts the strongest influence on sesLCBD changes among all considered

factors. These results provide robust evidence that U.S. freshwater watersheds with higher nonnative species proportions experience significantly greater LCBD reductions relative to historical states. The results confirmed our second hypothesis that nonnative species reduce LCBD across watersheds through the process of community homogenization. These findings corroborate the established understanding that biological invasions contribute to biodiversity loss (Villéger et al., 2011). The invasion process enables highly adaptable invasive species to spread across multiple watersheds, leading to convergent species composition among communities and thereby reducing taxonomic beta diversity through biotic homogenization (Olden et al., 2006; Villéger et al., 2011). Our results reveal a significant inverse relationship between total species richness and LCBD. This pattern is consistent with the concept of biotic homogenization, where the increased species richness associated with invasions corresponds to a decline in compositional uniqueness (Olden et al., 2008). Specifically, our data suggest that when total richness surpasses a certain threshold, it signifies a state of heightened community similarity across watersheds, which in turn diminishes the individual contribution of each watershed to regional beta diversity (Olden et al., 2004). The research by Su et al. (2023) demonstrates that watersheds become less susceptible to biological invasions as their functional originality. Consequently, ecosystems with higher functional originality consistently support fewer established nonnative species. Compared to watersheds with higher invasion levels, these ecosystems are better able to maintain the distinctiveness of their species composition. Thus, while other watersheds exhibit reduced contributions to beta diversity following invasion events, those with higher FOri demonstrate a more pronounced contribution to overall beta diversity. Additionally, the study observed that human activities, represented by GDP, exerted limited influence on diversity changes. This may reflect both the choice of socioeconomic indicators and the scale of analysis employed in the research (Dala-Corte et al., 2019).

Conservation implications and limitations

Based on the findings of this study and the ecological implications of LCBD (reflecting compositional uniqueness) and FOri (functional similarity), we propose a conservation framework targeting three types of communities to mitigate biotic homogenization and reduce invasion risk. Watersheds with high LCBD, often characterized by distinct species assemblages, are highly vulnerable to biological invasions and should be designated as a priority (Gavioli et al., 2019). Communities with low species richness yet high compositional distinctiveness possess fragile ecological foundations and significant uniqueness, requiring enhanced protection to prevent diversity loss. Additionally, communities with low FOri values exhibit high functional similarity among species and are likely to contain substantial unoccupied functional niches, creating favorable conditions for nonnative species establishment; thus, they also warrant increased conservation attention (Mouillot et al., 2013). The conservation framework derived from this analysis of U.S. freshwater fish communities offers transferable insights for managing fish diversity in other regions facing similar biological invasion pressures.

This study provides important insights into the effects of nonnative species on beta diversity across large scale watersheds in the U.S., yet it has several limitations. First, the reliance on two datasets from before and after invasion makes it difficult to fully capture the temporal dynamics of how nonnative species influence the contribution of each watershed to total beta diversity over time. Second, potential “effort bias” of data collection—where variations in sampling intensity across watersheds may have gone undetected—could affect the accuracy of estimated proportions of nonnative species in each watershed. Third, the analysis is based solely on presence-absence data, lacking quantitative information such as species abundance. Finally, although the BRT model identified drivers of changes in LCBD, spatial effects may potentially affect the interpretability of the statistical model and should be addressed in future research.

Conclusion

Based on macro-scale data across the U.S., this study examines the impact of nonnative species on LCBD in freshwater fish communities, providing critical evidence for informing biodiversity conservation policies. Our findings reveal that watersheds in the western and eastern U.S. exhibited significantly higher LCBD. However, invasions by nonnative species have substantially altered these original spatial patterns, underscoring the need for adaptive management strategies that incorporate dynamic ecological processes and enhance preventive measures against future invasions. Importantly, synergistic conservation of high-LCBD would not only help sustain community diversity but also support ecosystem functioning and associated services. Given the pronounced influence of nonnative species on high-LCBD watersheds, we recommend a dual-track conservation approach: intensifying protection in critical watersheds while implementing stricter regulations on human-mediated fish introductions. It should be noted, however, that our conclusions are derived from macro-scale data within the U.S., and the spatiotemporal scope of this study may not fully capture fine-grained invasion dynamics. Potential sampling biases and spatial autocorrelation in the dataset may also affect model interpretations. Future studies should integrate multi-scale monitoring networks, long-term temporal data, and advanced spatial modeling to better elucidate the complex ecological mechanisms underlying invasion impacts.

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CRediT authorship contribution statement

Congjun Xu: Writing – original draft, Visualization, Formal analysis, Conceptualization. **Jiping Li:** Writing – review & editing, Methodology. **Chenggong Du:** Methodology, Formal analysis. **Naisen Liu:** Writing – review & editing, Methodology, Conceptualization. **Jun Xu:** Writing – review & editing, Supervision, Conceptualization. **Wei Zhang:** Writing – review & editing, Formal analysis. **Guohuan Su:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.baae.2025.12.012.

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